



CS & IT ENGINEERING

Discrete Mathematics

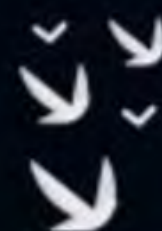
Mathematical Logic

Lecture No.- 05

By- Satish Sir



Recap of Previous Lecture



Topic

Logic Part-04



Topics to be Covered



Topic

Logic Part-05



predicate logic $\therefore \begin{matrix} \nearrow T \\ \rightarrow F \end{matrix}$

$P \wedge Q$

predicate variable.

$P(x)$: x is even no. (open stmt) $\rightarrow$ it is type of stmt

$x=0$ $P(0)$: 0 is even no. (T)

$x=1$. $P(1)$: 1 is even no. (F)

Such that we cannot define directly truth value, but once we put value of x , then we can define

* Domain / domain of discourse / universe / universe of discourse.
Set of possible choices for open stmt:



$x=0$

$x=1$

open
stmt:

x is even no.

0 is even no.

1 is even no.

$P(x)$: x is even no.

D:

<u>0 (T)</u>
1 (F)
2 (T)

D: $\{0, 1, 2\}$

$x=0$ 0 is even no. (T) ✓

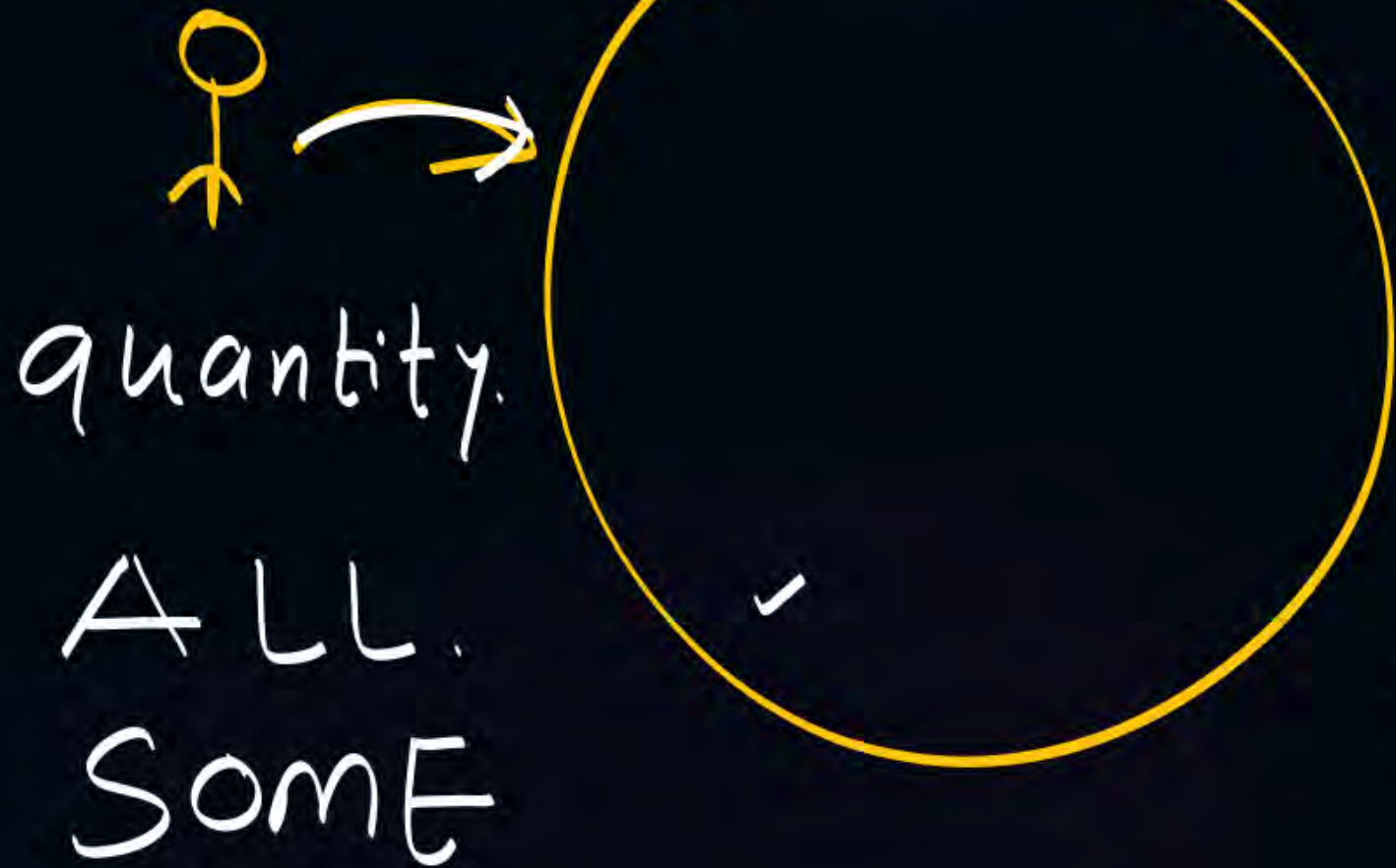
$x=1$ 1 is even no. (F)

$x=2$ 2 is even no. (T)

y

D:

• 0			
• F		T	
• T			• F
• T			



Quantifier:

mathematical tool, which is used in order to get truth value in terms of quantity.

ALL $\rightarrow$ universal quantifier.

$\forall x. P(x)$

for each x , such that $P(x)$.

for all value of x , it should satisfy $P(x)$.
 for all value of x , such that $P(x)$.
all of x , such that $P(x)$.

$$\underline{\forall x. \rightarrow \text{True.}}$$

when all elements.
are True in domain.



$$\underline{\forall x \rightarrow \text{False.}}$$

→ at least 1 element is false.



$D: \mathbb{Z}$

$$P(x): x^2 \geq 0$$

$$\forall x P(x)$$

OR

$$\underline{\forall x (x^2 \geq 0)}$$

→ True

$D: \mathbb{Z}$

$$P(x): x^2 > 0$$

$$\forall x P(x)$$

OR

$$\underline{\forall x (x^2 > 0)}$$

↪ false.

$x=0$
False.

Existential quantifier: $\exists x$.

$$\exists x P(x)$$

there exist at least 1 value of x , For that $P(x)$ is True.

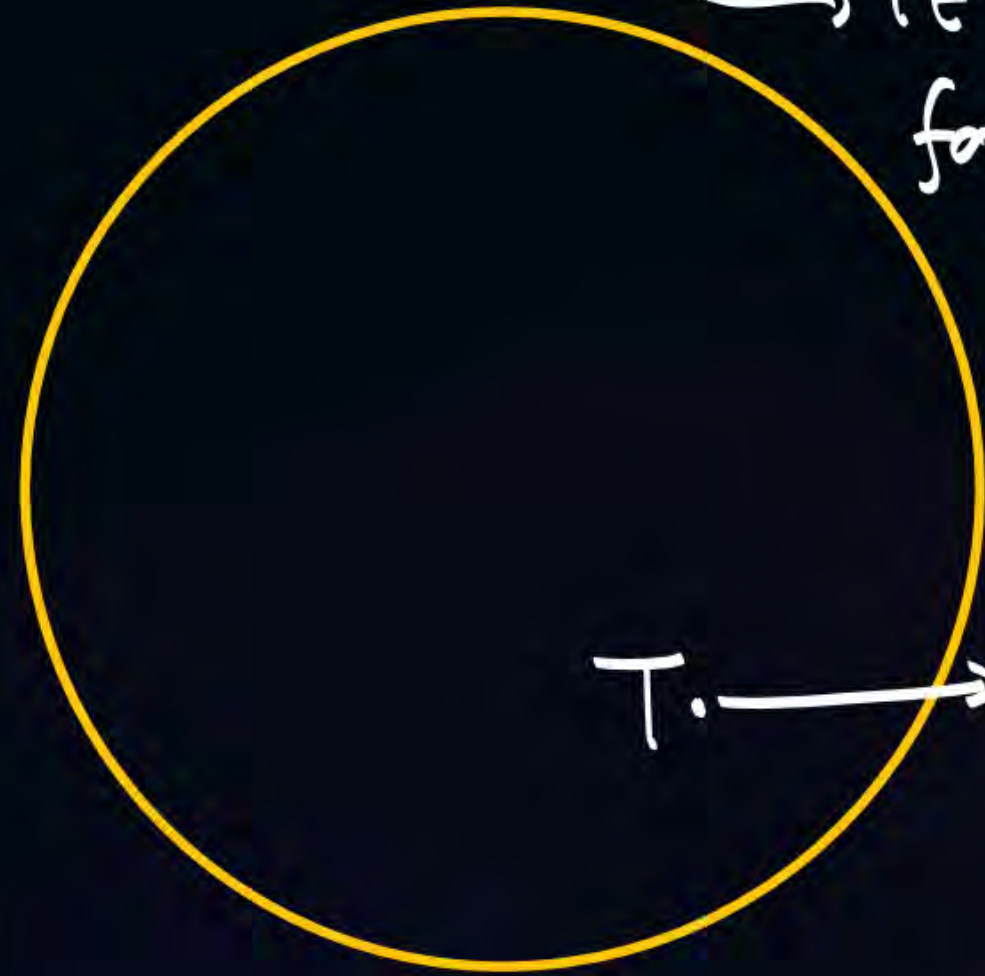
there exist x , such that $P(x)$.

some of x , such that $P(x)$.

at least x , such that $P(x)$

$\exists x \rightarrow \text{True.}$

→ it is true
for at least
1 element.



T. → at least
1 is True.

$\exists x \rightarrow \text{false.}$

→ when all
elements
are false.



$$D: \mathbb{Z}$$

$$p(x): x^2 = 4$$

$$\exists x p(x)$$

OR

$$\exists x (x^2 = 4)$$

→ True $x = 2$

$x = -2$

$$D: \{1, 2, 3\}$$

$$p(x): x^2 < 1$$

$$\exists x (x^2 < 1)$$

↪ false.

$\mathbb{D}: \mathbb{Z}$

$\{P(x): x^2 \geq 0\}$

$\forall x P(x) \rightarrow \exists x P(x)$

True



True

$\forall x P(x) \rightarrow \exists x P(x)$

OR

$\left\{ \begin{array}{l} \exists x P(x) \rightarrow \forall x P(x) \\ \text{True} \end{array} \right.$ False

Negation of quantifier

$D: \{1, 2, 3\}$

$\forall x P(x) \rightarrow \underline{\text{True}}$

$\exists x P(x)$

negate.

$$\neg \forall x P(x) \equiv \neg [P(1) \wedge P(2) \wedge P(3)]$$

$$\begin{aligned} \neg \forall x P(x) &\equiv \exists x \neg P(x) \\ \neg \exists x P(x) &\equiv \forall x \neg P(x) \end{aligned}$$

$$\equiv \neg P(1) \vee \neg P(2) \vee \neg P(3)$$

$$\equiv \exists x \neg P(x)$$

$$D: \{1, 2, 3\}.$$

$$\underline{P(x)}: x + 1 = 4.$$

$$x = 1 \quad 1 + 1 = 4 (F)$$

$$x = 2 \quad 2 + 1 = 4 (F)$$

$$x = 3 \quad 3 + 1 = 4 (T)$$

$$Q(x): x + 2 = 5.$$

$$x = 1 \quad 1 + 2 = 5 (F)$$

$$x = 2 \quad 2 + 2 = 5 (F)$$

$$x = 3 \quad 3 + 2 = 5 (T)$$

$$\exists x [P(x) \wedge Q(x)]$$

$D: \{1, 2, 3\}$
 $\exists x [P(x) \wedge Q(x)]$

$x=1$	$P(1) \wedge Q(1)$	$\rightarrow F$
$x=2$	$P(2) \wedge Q(2)$	F
$x=3$	$P(3) \wedge Q(3)$	T

T

$\exists x P(x)$		$\wedge$	$\exists x Q(x)$	
$P(1)$	F		$Q(1)$	F
$\vee$			$\vee$	
$P(2)$	F	$\wedge$	$Q(2)$	F
$\vee$			$\vee$	
$P(3)$	T		$Q(3)$	T
T			T	
T		$\wedge$	T	
T			T	

(valid) $\underline{\exists x [p(x) \wedge Q(x)]} \rightarrow \underline{\exists x p(x) \wedge \exists x Q(x)}$

T. check.

use inference
Rule

$$\exists x [P(x) \wedge Q(x)] \rightarrow \underbrace{\exists x P(x)}_{\text{True}} \wedge \underbrace{\exists x Q(x)}_{\text{True}} \equiv \text{True}$$

$$\exists x [p(x) \wedge Q(x)] \rightarrow \exists x p(x) \wedge \exists x Q(x) \text{ (valid)}$$

Check:

fill in the
blanks.



$$* \exists x [p(x)] \wedge \exists x Q(x) \rightarrow \exists x [p(x) \wedge Q(x)]$$

$$* \exists x [p(x) \vee Q(x)] \quad \exists x p(x) \vee \exists x Q(x)$$

$$\forall x [p(x) \wedge Q(x)] \quad \forall x p(x) \wedge \forall x Q(x)$$

$$\forall x [p(x) \vee Q(x)] \quad \forall x p(x) \vee \forall x Q(x)$$

$$\forall x [p(x) \rightarrow Q(x)] \quad \forall x p(x) \rightarrow \forall x Q(x)$$



2 mins Summary



Topic

One

Logic Part-04

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU